Report Nº 3
Team Out of Lane
Date 17.02.2025

1 Introduction

This report details the activities carried out to date in the third sprint. These activities emphasized lane detection and lane keeping.

2 Planned activities

Category	*Activity*	*Responsible*
Hardware improvements	Mechanical calibration	Alexander P.
	Research GPU addition	Alexander P. and Evelyn C.
	Research hardware additions (headlight)	Alexander P., Carolina N. and Evelyn C.
	Define other necessary sensors, define use-case, integration (IMU, distance), preprocessing, noise cancelling	Alexander P., Carolina N. and Evelyn C.
Sensing and input package	Define use-cases and test given servers information (localization on map, cars interaction, GPS interaction, interaction with objects)	Alexander P., Carolina N., Evelyn C. and Pablo F.
Computer vision package (RPi)	Implement lane detection	Alexander P. and Pablo F.
	Intersection detection	Alexander P. and Pablo F.
	Implement traffic light detection	Evelyn C.
Project Documentation	Architecture revision	Pablo F.
	Architecture diagram	Evelyn C. and Pablo F.
	Prepare traffic sign dataset	Alexander P., and Evelyn C.
	Define decision making (priorities of actions and state flow)	Alexander P., Carolina N., Evelyn C. and Pablo F.
	Document the 3rd report	Carolina N. and Pablo F.

3 Status of planned activities

Activity	*%*	*Observations*
Mechanical calibration	75%	We decided to focus on other activities and leave this small correction in the next sprint.
Research GPU addition	100%	We have decided we are going to use a Jetson Nano Orin. Currently we are working on its acquisition.
Research hardware additions (headlight)	80%	The ultrasound sensors are installed and working. The lights are almost ready, but we decided to focus on tidying up other aspects and leave the lights for the next report.
Define other necessary sensors, define use-case, integration (IMU, distance), preprocessing, noise cancelling	25%	We considered different possible scenarios and recorded the car's reactions to improve the algorithm. This activity is planned to last for two sprints to refine the car's reaction.
Define use-cases and test given servers information	5%	We began investigating the data that will be used and potential interactions and behaviors. This activity is planned to last for two reports to refine the car's reaction.
Implement lane detection	70%	We have developed a lane detection algorithm using OpenCV, which can follow the lane markings and stay within the designated lanes. We are currently refining the detection of crosswalks and implementing a better controller.

Activity	*%*	*Observations*
Implement traffic sign detection	50%	We integrated YOLO and the RPi. The car is capable of recognizing traffic signs.
Intersection detection	10%	We have started working theoretically on the detection of intersections and how to proceed when the car is at one.
Implement traffic light detection	10%	We have started working theoretically with the traffic light detector.
Architecture revision	100%	It was not necessary to modify our architecture
Architecture diagram	100%	It was not necessary to modify our architecture diagrams.
Prepare traffic sign dataset	75%	We printed the traffic signs, and the dataset is almost ready.
Define decision making	25%	We are currently working on a lane keeping controller. A decision-making process was already implemented.
Document the 3rd report	100%	The report itself.

4 General status of project

The team has advanced in its algorithmic approach to lane detection and lane keeping. The car can stay on track on straight lanes and on curves, in any direction.
On the other hand, the Brain is now capable of detecting traffic signs and has already implemented some logic on the sight of the "stop" signal.

5 Upcoming activities

To be continued from the 3rd sprint

- **Define other necessary sensors, define use-case, integration (IMU, distance), preprocessing, noise cancelling**
- **Define use-cases and test given servers information (localization on map, cars interaction, GPS interaction, interaction with objects)**
- **Mechanical calibration.**
- **Research hardware additions (headlight).**
- **Implement lane detection.**
- **Implement traffic sign detection.**
- **Intersection detection.**
- **Implement traffic light detection.**
- **Prepare traffic sign dataset.**
- **Define decision making (priorities of actions and state flow).**

New activities

- **Testing sensors and actuators for proper capture, processing and response.**
- **Object detection & classification.**
- **Environmental server interaction.**
- **Traffic light detection and classification.**
- **Define robustness and safety measures.**
- **Architecture revision.**
- **Architecture diagram.**
- **Document the 4th report.**